

Create or Modify System Process

Sub-techniques (5)



Adversaries may create or modify system-level processes to repeatedly execute malicious payloads as part of persistence. When operating systems boot up, they can start processes that perform background system functions. On Windows and Linux, these system processes are referred to as services.^[1] On macOS, launchd processes known as [Launch Daemon](#) and [Launch Agent](#) are run to finish system initialization and load user specific parameters.^[2]

Adversaries may install new services, daemons, or agents that can be configured to execute at startup or a repeatable interval in order to establish persistence. Similarly, adversaries may modify existing services, daemons, or agents to achieve the same effect.

Services, daemons, or agents may be created with administrator privileges but executed under root/SYSTEM privileges. Adversaries may leverage this functionality to create or modify system processes in order to escalate privileges.^[3]

ID: T1543

Sub-techniques: [T1543.001](#), [T1543.002](#), [T1543.003](#), [T1543.004](#), [T1543.005](#)

① Tactics: [Persistence](#), [Privilege Escalation](#)

① Platforms: Containers, Linux, Windows, macOS

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[Version Permalink](#)

Procedure Examples

ID	Name	Description
S1194	Akira_v2	Akira_v2 can create a child process for encryption. ^[4]
S1184	BOLDMOVE	BOLDMOVE can free all resources and terminate itself on victim machines. ^[5]
S9015	BRICKSTORM	BRICKSTORM has created a new background session and has spawned a child process of a parent process when it determines it is not running in its intended state. ^[6]
S9042	CanisterWorm	CanisterWorm can establish persistence in CI/CD environments by launching a background process (deploy.js) with stolen tokens. ^[7]
S0401	Exaramel for Linux	Exaramel for Linux has a hardcoded location that it uses to achieve persistence if the startup system is Upstart or System V and it is running as root. ^[8]
S1152	IMAPLoader	IMAPLoader modifies Windows tasks on the victim machine to reference a retrieved PE file through a path modification. ^[9]
S1121	LITTLELAMB.WOOLTEA	LITTLELAMB.WOOLTEA can initialize itself as a daemon to run persistently in the background. ^[10]
S1142	LunarMail	LunarMail can create an arbitrary process with a specified command line and redirect its output to a staging directory. ^[11]

Mitigations

ID	Mitigation	Description
M1047	Audit	Use auditing tools capable of detecting privilege and service abuse opportunities on systems within an enterprise and correct them.
M1040	Behavior Prevention on Endpoint	On Windows 10, enable Attack Surface Reduction (ASR) rules to prevent an application from writing a signed vulnerable driver to the system. ^[12] On Windows 10 and 11, enable Microsoft Vulnerable Driver Blocklist to assist in hardening against third party-developed drivers. ^[13]
M1045	Code Signing	Enforce registration and execution of only legitimately signed service drivers where possible.
M1033	Limit Software Installation	Restrict software installation to trusted repositories only and be cautious of orphaned software packages.
M1028	Operating System Configuration	Ensure that Driver Signature Enforcement is enabled to restrict unsigned drivers from being installed.
M1026	Privileged Account Management	Manage the creation, modification, use, and permissions associated to privileged accounts, including SYSTEM and root.
M1022	Restrict File and Directory Permissions	Restrict read/write access to system-level process files to only select privileged users who have a legitimate need to manage system services.
M1054	Software Configuration	Where possible, consider enforcing the use of container services in rootless mode to limit the possibility of privilege escalation or malicious effects on the host running the container.
M1018	User Account Management	Limit privileges of user accounts and groups so that only authorized administrators can interact with system-level process changes and service configurations.

Detection Strategy

ID	Name	Analytic ID	Analytic Description
DET0571	Detection of System Process Creation or Modification Across Platforms	AN1575	Detects command-line or API-based creation/modification of Windows Services via <code>sc.exe</code> , <code>powershell.exe</code> , <code>services.exe</code> , or <code>ChangeServiceConfig</code> . Looks for creation/modification of autostart services via registry changes, file drops to <code>System32\services</code> , and anomalous parent-child process trees.
		AN1576	Detects creation or modification of <code>systemd</code> service units, addition of cron jobs that invoke binaries on boot, or suspicious writes to <code>/etc/init.d/</code> . Monitors <code>chmod +x</code> and <code>systemctl</code> execution paths, especially from non-root parent processes.
		AN1577	Detects creation or modification of <code>LaunchDaemon</code> or <code>LaunchAgent</code> plist files under <code>/Library/LaunchDaemons/</code> , <code>~/Library/LaunchAgents/</code> , or similar. Monitors execution of <code>launchctl</code> , property list edits, and file permission changes.
		AN1578	Detects creation of new container system processes via <code>docker run --restart</code> , <code>kubectrl exec</code> to init containers, or modification of container init specs. Flags container images that override entrypoints to embed persistence behaviors.

References

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6. [DHS/CISA. \(2026, February 11\). AR25-338A: BRICKSTORM Backdoor. Retrieved April 16, 2026.](#)
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9. [PwC Threat Intelligence. \(2023, October 25\). Yellow Liderc ships its scripts and delivers IMAPLoader malware. Retrieved August 14, 2024.](#)
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13. [Jordan Geurten et al. . \(2022, March 29\). Microsoft recommended driver block rules. Retrieved April 7, 2022.](#)